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Position estimation of synchronous electrostatic film motors under pulse-voltage operation by using driving currents

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ABSTRACT

Synchronous electrostatic film motors under pulse-voltage operation require position estimation, as it is unavoidable to step out from synchronization due to unexpected large external disturbance. However, the existing position estimation method using driving currents can only work well under sine-voltage operation and fails with a huge error for the pulse-voltage operation due to the impulsive currents at the switching time of the pulse-voltage. This work proposes a new position estimation method dedicated for the pulse-voltage operation, which is mainly used to track the slider position at out-of-step. The impulse currents is removed directly in this method, however, the removal period affects the estimation performance since some essential information of position can also be removed. Theoretically, as long as the information loss is less than a certain level, the proposed method can always work well. Simulation results confirmed that the slider position can be estimated correctly with a resolution of an electrode pitch if the duration of the impulsive currents is short. Experiments under various conditions were carried out by using a real motor, and the results verified that the proposed method can not only successfully track the slider position at out-of-step, but also during synchronous operation.

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1. Introduction

Electrostatic motor was invented in the 1740s by Benjamin Franklin, which is much earlier than the first magnetic machine conceived by Michael Faraday in the 1820s [1,2]. In the past, electrostatic motors were conceived purely for research purposes due to their low performance compared with magnetic motors, even after it began to be used in industrial utility in 1933 [3]. The electrostatic motors have never caught up on the magnetic motors until around 1990, when it revealed excellent performance in micro-scale structures [4–7]. Since then, electrostatic motor began to prosper in micro-electromechanical system (MEMS) [8–10] and micro-robot systems [11–14,15]. At the same time, new configurations of macro-scale linear film motors with high power density were constructed, in which electrodes were arranged in parallel and printed on flexible plastic substrates or thin boards [16–18]. In [17], the power per weight ratio of the electrostatic film motor was reached up to 230 W/kg by applying high voltage (around 2 kV) when it was operated in dielectric fluid. By stacking 50 pairs of electrode films, the motor prototype was able to generate a large thrust force up to 310 N [19]. Recently, electrostatic rotating motors competitive with magnetic machinery have also been conceived [20–23–26,27]. In [20], a voltage-induction synchronous rotational electrostatic motor was proposed, in which the rotor electrodes are charged indirectly to eliminate the wiring system. In [21–24], the cylindrical electrodes were arranged in the axial direction to increase the effective electrode area, and the motors provided high performance in dielectric fluid. In [25–27], a prototype electrostatic motor was fabricated by using printed circuit board technology, and its demonstrated torque densities were comparable to surface permanent magnet machines. In future, the electrostatic motors will be potentially competitive with, or even superior to, magnetic machines in some specific applications [28].

The electrostatic film motors stand out from various electrostatic motors due to their unique characteristics, such as thin structure, mechanical flexibility, high power density, selectable optical transparency and non-magnetic property. Various operation principles have been developed for the film motors during past decades, including synchronous types [17,18] and induction types [29,30], among which this work focuses on the synchronous electrostatic film motor. The electrostatic film motors of synchronous type have potential to be used in robotic systems [31,32], computer-human-

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Nomenclature

$\mathbf{V}$	Applied voltage vector, [V]
$\mathbf{C}$	Capacitance matrix of electrostatic film motor, [F]
$\mathbf{C}_0$	Constant component of $\mathbf{C}$, [F]
$\mathbf{C}_{\theta_x}$	Position-dependent variable component of $\mathbf{C}$, [F]
x, θ_x	Slider position and its electric angle expression, [m], [rad]
$\mathbf{I}$	Driving currents of electrostatic film motor, [A]
$\mathbf{I}_p$	Plateau currents in $\mathbf{I}$, [A]
$\mathbf{I}_s$	Switching currents in $\mathbf{I}$, [A]
$\mathbf{I}_{\alpha\beta}$	Simplified driving currents after Clarke transformation, [A]
$\mathbf{M}_{\alpha\beta}$	Transfer matrix for Clarke transformation

interaction (CHI) systems [33,34], and within magnetic resonance (MR) environments [35,36]. The performance of the electrostatic film motors can be increased by stacking many pairs of films; however, only one pair of films is used in light applications, such as CHI systems. In any practical applications, the motors possibly step out from synchronous motion due to severe environmental disturbances, especially in case of light applications due to the low performance of single motor. Therefore, position estimation is required for recovery from the position errors. Although position is typically measured by using a position sensor/encoder, a simple and low-cost system without spoiling the unique characteristics of the electrostatic film motors is expected for synchronous-type motors.

Meanwhile, many exclusive position sensing methods that are suitable to electrostatic film motors were developed in previous research. In [37], a position sensor using the same structure as the electrostatic film motor was proposed, which could realize a high resolution by using high frequency sensing signals. It can be fabricated on the same electrode film as the motor. However, it requires an extra area in the film, which reduces the area for actuation and thus the motor performance is compromised. Therefore, another self-sensing method based on signal injection was proposed for the synchronous electrostatic motors to eliminate the sensor electrodes [38–40]. Similarly with the self-sensing method in electromagnetic machines [41–44], high frequency voltage signals were injected to the actuation electrodes of the electrostatic motors, and the slider/rotor position information was extracted from the resultant currents. However, these self-sensing methods require complicated systems for injecting/extracting high frequency signals to/from the driving signals.

To realize position estimation with a simpler setup, a new self-sensing method was proposed for the synchronous electrostatic film motors [45]. Similarly with the self-sensing method of electromagnetic machine/generator in [46,47], the slider position information is also extracted from the driving currents. Its theoretical framework was derived under sine-wave operation, where a three-phase sinusoidal high voltage is used as a driving voltage signal. The resulting driving currents are also sinusoidal waves, with some distortion caused by the slider motion. The previous method extracted the distortion from the sinusoidal wave by using Park transformation, through which the slider position was estimated. Since no high frequency sensing signal is required, the resulting measurement system becomes very simple and low cost.

On the other hand, pulse-voltage is also widely used to operate the electrostatic motors. Compared with operation under sinusoidal voltage, the pulse-voltage operation is attractive and expected to be used in various applications due to its simplicity and low cost. In this case, impulse currents are created in the driving currents at the switching time of the pulse-voltage, which can not be handled properly by the previous method [48]. As a result, the position estimation failed with huge errors at the switching time. Furthermore, the impulse currents and position estimation under pulse-voltage operation have not been discussed in other research [38–40]. Therefore, this work proposes a position estimation method for the synchronous electrostatic film motors under the pulse-voltage operation, in which the slider position (especially at out-of-step) can be estimated correctly by eliminating the influence of the resultant impulse currents. Although the proposed method is derived by using the model of the electrostatic film motor, it is also suitable for rotational electrostatic motors due to the same underlying principle.

The rest of this paper is structured as follows: Section 2 briefly introduces the synchronous electrostatic film motor with pulse-voltage operation and proposes the position estimation method using the driving currents. The driving currents under the pulse-voltage operation are very different from those in sine-wave operation due to the impulse currents. Therefore, the driving currents are theoretically derived and discussed. Then, the position estimation method using the driving currents for pulse-voltage operation is proposed, in which the impulse currents are removed. The removal of impulse currents poses a new problem in the signal processing, in particular for phase-unwrapping process, which is discussed in the last part of Section 2. Section 3 investigates the suitable conditions for the position estimation method through various simulations. Section 4 shows the results and discussions in verification experiments under various conditions. Last section is conclusion.

2. Position estimation method

2.1. Electrostatic film motors

The basic structure and the pulse-voltage operation mechanism of the synchronous electrostatic film motor are shown in Fig. 1. It consists of two films, in which 3-phase fine-pattern electrodes are arranged typically using flexible printed circuit technology; one is the slider while the other one is the stator. In the synchronous electrostatic film motor, any pair of electrodes can be regarded as a capacitor, as shown in Fig. 1. When pulse voltages $[+V_0, -V_0, 0]$ with a certain sequence are applied to the electrodes, charges are excited to generate coulomb forces between the two films. The horizontal components of the coulomb forces propel the slider, as shown in Fig. 1. As a result, the electrostatic film motor acts as a stepper motor. Differently from the sine-voltage operation, which can produce smooth synchronized motion [18], the pulse-voltage operation only generates stepwise motion due to the thrust force ripple. There are several 3-phase voltage sequences that can be used for pulse-voltage operation. The sequence used in this work is shown in Fig. 2. The voltage of each phase changes in the sequence

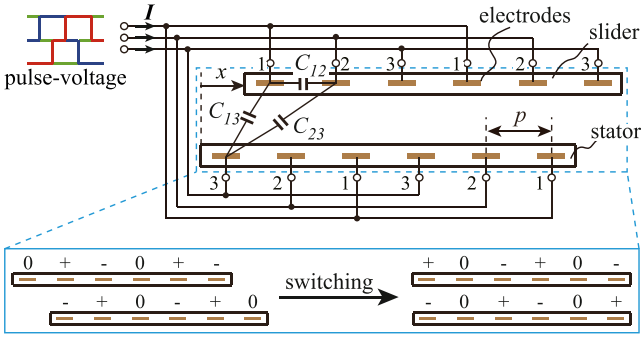


Fig. 1. Basic structure and driving principle of synchronous electrostatic film motor under pulse-voltage operation.

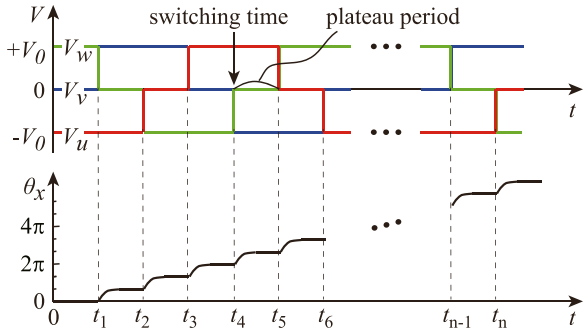


Fig. 2. One operation type of the 3-phase pulse voltages. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

[+V_0, +V_0,0, −V_0, −V_0,0]. At each switching time, the slider film is moved by one electrode pitch p. If external disturbance that is larger than the maximum thrust force of the motor was added to the slider, it will be forced to step out from the stepwise/synchronized motion.

The thrust force between the two films is described as

$$F_{thrust} = \frac{1}{2}\mathbf{V}^T \frac{d\mathbf{C}}{dx}\mathbf{V} \tag{1}$$

where $\mathbf{V}$ is the applied voltage vector, and $\mathbf{C}$ is the capacitance matrix of the motor [18]. The capacitance matrix can be described as

$$\mathbf{C} = \mathbf{C}_0 + \mathbf{C}_{\theta_x} \tag{2}$$

where $\mathbf{C}_0$ is a constant matrix

$$\mathbf{C}_0 = \begin{Bmatrix} C_{11}(=C_{22}=C_{33}) = C_{st0} + C_{sl0} - 2C_{m0} \\ C_{12}(=C_{13}=C_{23}) = -C_{t0} - C_{l0} - 2C_{m0} \end{Bmatrix}$$

and $\mathbf{C}_{\theta_x}$ is a position-dependent variable matrix

$$\mathbf{C}_{\theta_x} = \begin{Bmatrix} C_{11}(=C_{23}) = -2C_{m1}\cos(\theta_x - 2\pi/3) - 2C_{m2}\cos(2\theta_x + 2\pi/3) \\ C_{22}(=C_{13}) = -2C_{m1}\cos(\theta_x) - 2C_{m2}\cos(2\theta_x) \\ C_{33}(=C_{12}) = -2C_{m1}\cos(\theta_x + 2\pi/3) - 2C_{m2}\cos(2\theta_x - 2\pi/3) \end{Bmatrix}$$

in which C_{st0}, C_{sl0}, C_{t0}, C_{l0}, C_{m0}, C_{m1} and C_{m2} are all positive constants [45]. Here, θ_x is the electric angle expression of the slider position x, and satisfies θ_x=2πx/3p.

2.2. Driving currents

The driving currents under the pulse-voltage operation is discussed in this part. Differently from sine-voltage, pulse-voltage is discontinuous and changes suddenly at the switching time t_n, as shown in Fig. 2. As a result, impulse currents occur at the switching time. Therefore if the position self-sensing method for the sine-wave operation in [45] is simply applied to the pulse-voltage operation, the impulse currents disrupt the position estimation [48]. For proper estimation, the expression of the driving currents under the pulse-voltage operation is derived in this section by referring to the previous method in [45].

The 3-phase pulse-voltage vector is defined as

$$\mathbf{V} = [V_u, V_v, V_w]^T \tag{3}$$

where, V_u, V_v, V_w are the voltages applied to the u-, v-, and w-phase electrodes, respectively. The pulse-voltage is switched at t_n (n=1, 2, …) and is kept at a constant voltage during the plateau period between two adjacent switching times. The driving currents flowing in the motor can be calculated as

$$\mathbf{I} = \frac{d\mathbf{Q}}{dt} = \frac{d(\mathbf{CV})}{dt} \tag{4}$$

where the vector $\mathbf{Q}$ represents charges accumulated in the electrodes of the three phases, and is $\mathbf{Q}$=$\mathbf{CV}$.

For simplicity, the constant C_{m2} in Eq. (2) is neglected in the following theoretical calculation. From Eqs. (2) and (4), the driving currents are calculated as

$$\mathbf{I} = \frac{d(\mathbf{CV})}{dt} = \underbrace{\frac{d\mathbf{C}}{dt}\mathbf{V}}_{\mathbf{I}_p} + \underbrace{\mathbf{C}\frac{d\mathbf{V}}{dt}}_{\mathbf{I}_s} \rightarrow \begin{Bmatrix} \mathbf{I}_p = \frac{d\mathbf{C}}{dt}\mathbf{V} \\ \mathbf{I}_s = \mathbf{C}\frac{d\mathbf{V}}{dt} \end{Bmatrix} \tag{5}$$

The first term of the current is caused by the change of the capacitance matrix, which is a function of the slider position. Therefore, the first term equals zero when there is no slider motion. During the infinitesimally short switching period, we can assume that the slider does not move and this term vanishes. Since the slider moves during the plateau period, the first term appears only in the plateau period. Therefore, this term will be called “plateau currents”, $\mathbf{I}_p$.

On the other hand, the second term occurs only at the voltage switching time, and thus is called “switching currents”, $\mathbf{I}_s$. The switching currents $\mathbf{I}_s = [i_{su}, i_{sv}, i_{sw}]^T$ become impulse signals and can be expressed as

$$\mathbf{I}_s = \mathbf{C}\frac{d\mathbf{V}}{dt} = \mathbf{C}\delta(t - t_n)[\mathbf{V}(t_n^+) - \mathbf{V}(t_n^-)] \tag{6}$$

where $\delta(t)$ represents the unit impulse function, t_n is the switching time, and t_n^+ and t_n^- mean the instantaneous times after and before t_n.

Since the plateau currents are caused by the slider motion, we can extract the slider position by analyzing the plateau current. Using Eq. (2), $\mathbf{I}_p$ can be simplified as

$$\mathbf{I}_p = \frac{d\left(\mathbf{C}_0 + \mathbf{C}_{\theta_x}\right)}{dt}\mathbf{V} = \frac{d\mathbf{C}_{\theta_x}}{dt}\mathbf{V} \tag{7}$$

substituting the matrix component and the voltage in (3), $\mathbf{I}_p$ is obtained as

$$\mathbf{I}_p = \begin{bmatrix} i_{pu} \\ i_{pv} \\ i_{pw} \end{bmatrix} = 2C_{m1}\dot{\theta_x} \begin{bmatrix} \sqrt{3}(V_v - V_u)\cos\theta_x + 3V_w\sin\theta_x \\ \sqrt{3}(V_u - V_w)\cos\theta_x + 3V_v\sin\theta_x \\ \sqrt{3}(V_w - V_v)\cos\theta_x + 3V_u\sin\theta_x \end{bmatrix} \tag{8}$$

this equation clearly shows that the information of the slider motion is recorded in the plateau currents $\mathbf{I}_p$.

2.3. Decoding displacement by removing impulse currents

To estimate the slider position, the plateau currents $\mathbf{I}_p$ alone needs to be extracted from the driving currents. This requires removal of the switching currents $\mathbf{I}_s$. As the occurrence time of $\mathbf{I}_s$ is easy to be identified, this work will simply cut the driving current during the switching period, which has certain time length in reality. Considering the continuity of remained currents and possible non-zero calculation in the following, the switching currents $\mathbf{I}_s$ is replaced using the driving currents at t_n^-, as shown in Fig. 3. Thereafter, the remained currents is regarded as the plateau currents $\mathbf{I}_p$.

Clarke transformation is used to handle $\mathbf{I}_p$, as it can transfer the three-phase signal into a simpler 2-phase expression as

$$\mathbf{I}_{\alpha\beta} = \mathbf{M}_{\alpha\beta}\mathbf{I}_p \tag{9}$$

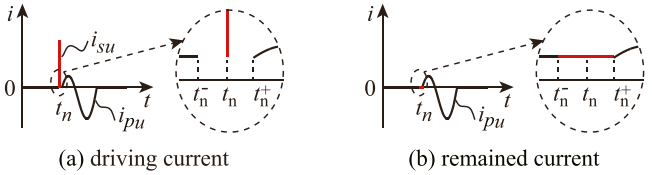


Fig. 3. Schematic diagram for removing switching current.

where

$$\mathbf{M}_{\alpha\beta} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{3}} & 0 \end{bmatrix}$$

is the transfer matrix. By using Eq. (8), the details of $\mathbf{I}_{\alpha\beta}$ are obtained as

$$\begin{aligned} \mathbf{I}_{\alpha\beta} &= \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} = 2C_{m1}\dot{\theta}_x \begin{bmatrix} \sqrt{3}(V_v - V_u)\cos\theta_x + 3V_w \sin\theta_x \\ \sqrt{3}(V_v - V_u)\sin\theta_x - 3V_w \cos\theta_x \end{bmatrix} \\ &= D\begin{bmatrix} \cos(\theta_x - \lambda) \\ \sin(\theta_x - \lambda) \end{bmatrix} \end{aligned} \tag{10}$$

where $D = 2\sqrt{3}C_{m1}\dot{\theta}_x\sqrt{(V_u - V_v)^2 + 3V_w^2}$ has a relation with the slider velocity $\dot{\theta}_x$ and the applied voltage. The phase λ depends on the pulse voltage, which only changes at the switching time, and is calculated as

$$\lambda = \arctan[-\sqrt{3}V_w/(V_u - V_v)] \tag{11}$$

by defining $\gamma = \theta_x - \lambda$, the expression of $\mathbf{I}_{\alpha\beta}$ can be simplified as

$$\mathbf{I}_{\alpha\beta} = D\begin{bmatrix} \cos\gamma \\ \sin\gamma \end{bmatrix} \tag{12}$$

in this expression, γ can be easily obtained as

$$\gamma = \arctan\frac{i_\beta}{i_\alpha} \tag{13}$$

then, the slider position is calculated by

$$x = \frac{3p}{2\pi}\theta_x = \frac{3p}{2\pi}(\gamma + \lambda) \tag{14}$$

In reality, the phase γ and λ can take any value within $(-\infty, \infty)$, while the arctan function can only be calculated in the range of $(-\pi/2, \pi/2)$. In order to get the correct values in signal processing, the calculated γ and λ need to be phase-unwrapped by using an phase unwrapping algorithm [49,50], in which a threshold value is required to judge whether the phase jump needs to be rectified.

Although the position estimation method discussed above and the previous position self-sensing method in [45] share the same concept, they are suitable for different operation conditions due to their different processing methods. The plateau currents and the switching currents in Eq. (5) correspond to "feature currents" and "base currents" under the sine-voltage operation in [45]. For both methods, the slider position is estimated from the plateau currents or the feature currents. Their difference depends on how to remove the switching currents or the base currents, which are impulse currents and sine-wave currents, respectively. In the previous method, the driving currents were simplified and transformed by using the Clarke and Park transformations. As a result, the sine-wave currents were turned into a constant, and thus the feature currents were distilled easily by subtracting the constant. However, the previous method had no means to deal with the impulse currents correctly, and failed to track the slider position under the pulse-voltage operation [48]. On the other hand, the impulse currents are removed directly from the driving currents in this work, and then the remained currents or the plateau currents are simplified by using the Clarke transformation. Finally, the position information is extracted from the simplified plateau currents. Compared with the previous method, the proposed method in this work has simple and easy signal processing, as Park transformation is virtually replaced by Eq. 11. Therefore, the position estimation method in this work is suitable for the pulse-voltage operation, while the previous method can only work well under sine-voltage operation.

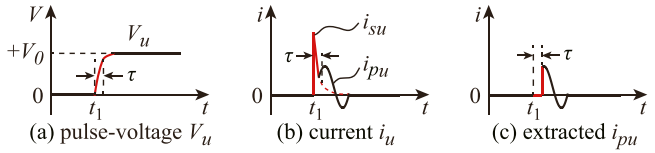


Fig. 4. Removing switching currents under practical condition. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

2.4. Feasibility under real pulse-voltage

Removing the switching currents $\mathbf{I}_s$ plays an important role in the proposed position estimation method. However, the switching currents is not easy to be removed in practice, as shown in Fig. 4, taking u-phase driving voltage/current as an example. In reality, pulse-voltage V_u applied to the motor has a rise time τ, which is defined as the time taken by the pulse-voltage to change from zero to 90% of the amplitude in this work. Then, the impulse current i_{su} turns into a signal that continues to decay over time, as shown in Fig. 4(b) using red lines. As a result, the plateau current i_{pu} mixes together with the switching current i_{su}. Considering the practicality, τ is used as the removal period of the switching current. Then, the removal of switching currents is expressed as

$$\mathbf{I}(t) = \mathbf{I}(t_n^-),\ t \in [t_n,\ t_n + \tau] \tag{15}$$

when the impulse current i_{su} during time τ is removed, some part of the plateau current i_{pu} will also be eliminated, as shown in Fig. 4(c). This causes information loss in position estimation.

Since the information loss can not be avoided, the requirement for successful position estimation needs to be clarified. Removing the switching currents by using Eq. (15) possibly causes discontinuity at point $t = t_n + \tau$, which will remain after the Clarke transformation and cause discontinuity in phase γ. This causes a problem in phase unwrapping for the arctan output. Discontinuity caused by the removal and discontinuity (jump of π) caused by the arctan function exist at the same time, as shown in Fig. 5. These two discontinuities need to be correctly identified.

For the pulse-voltage operation, γ changes with a step of $\pi/3$, which will be confirmed in the next section. Discontinuity caused in the calculation of arctan function is expressed by using blue dashed line, which is called as "π-jump". On the other hand, red dashed line is the discontinuity caused by removing the switching currents, which equals to the phase loss of the plateau currents $\Delta\gamma_\tau$ and is

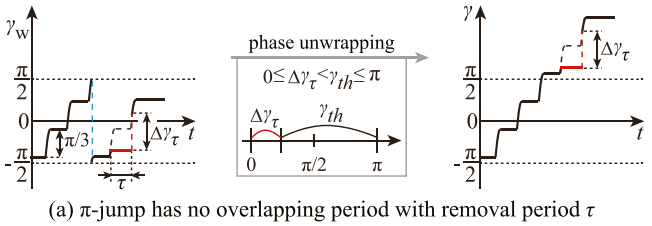


Fig. 5. Problems existing during phase-unwrapping. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

called as "R-jump". Only "π-jump" should be rectified by using phase unwrapping algorithm to get the correct phase γ. The requirement for successful phase unwrapping can be discussed under two conditions:

Condition 1: "π-jump" occurs outside the removal period τ, as shown in Fig. 5(a). The relation between "R-jump", "π-jump" and threshold value in phase unwrapping algorithm γ_{th} should be

$$0 \le \Delta\gamma_\tau < \gamma_{th} \le \pi \tag{16}$$

Condition 2: "π-jump" occurs during the removal period τ, as shown in Fig. 5(b). In "R-jump", $\Delta\gamma_\tau$ is divided into two components $\Delta\gamma_1$ and $\Delta\gamma_2$ by "π-jump". Then, "R-jump", "π-jump" and γ_{th} should satisfy the following relation.

$$\left\{\begin{matrix} 0 \le \Delta\gamma_\tau < \gamma_{th} \le \pi \\ \pi - \Delta\gamma_\tau \ge \gamma_{th} \end{matrix}\right\} \tag{17}$$

From Eqs. (16) and (17), the requirement for successful estimation under any conditions is derived as

$$\left\{\begin{matrix} 0 \le \Delta\gamma_\tau < \pi/2 \\ \Delta\gamma_\tau < \gamma_{th} \le \pi - \Delta\gamma_\tau \end{matrix}\right\} \tag{18}$$

which implies that the wrapped γ can be rectified correctly as long as "R-jump" caused by the removal is smaller than $\pi/2$. In the following sections, γ_{th} is set as $\pi/2$ to satisfy the above requirement.

3. Simulation

3.1. Block diagram

In the previous section, the position estimation method for pulse-voltage operation was proposed in theory by using the simplified model in which the constant C_{m2} in Eq. (2) was neglected. To verify whether this method can work well or not for the model of the real motor, numerical simulations were carried out on Simulink/Matlab 2019a.

The block diagram of the numerical simulation is shown in Fig. 6. Same with in [45], the area enclosed by red dashed lines describes the module of the electrostatic film motors. The enclosed area formed by blue dashed lines was used to simulate the motion dynamics of the motor. Differently from the previous work, the model was adjusted for the pulse-voltage operation. A set of pulse-voltage is fed to the motor block to obtain the thrust force and the driving currents by using Eq. (1) and (4). The ideal pulse-voltages $\mathbf{V}_{ideal}$ expresses an ideal pulse voltage, which is shown in Fig. 2. Considering the resistance of power supply in reality, an resistance R is set in the feedback loop to obtain driving voltages $\mathbf{V}$ with finite rise time τ. The resulting voltages $\mathbf{V}$ and the driving currents $\mathbf{I}$ are fed to the proposed method, which is enclosed by using green dashed lines, for estimating the slider position x_{cal}. By removing the impulse currents, the plateau currents $\mathbf{I}_p$ is obtained and then is simplified by using the Clarke transformation. The phase γ is calculated by using Eq. (13), while the phase λ is obtained by using Eq. (11) from the driving voltages. Finally, the slider position is estimated by using Eq. (14).

An external disturbance which is expressed as an impulsive signal whose amplitude is 10 N with a duration of 4 ms, was given to the slider to force the motor step out. The disturbance was added to the slider in two cases, during the plateau period and at a switching time. All the parameters of the motor used in the simulations, such as the capacitance coefficients, the slider mass, the electrode pitch and the viscous damping coefficient of the dielectric fluid, were adopted from the "regular motor" in [45]. Meanwhile, the amplitude of the pulse-voltage $\mathbf{V}_{ideal}$ was ± 1.5 kV and its switching frequency was 30 Hz. The sampling frequency in Simulink/Matlab was set as 10 kHz under the fixed-step 5th-order Dormand-Prince method.

3.2. Results

When the driving voltages $\mathbf{V}$ with a rise time of τ=0.3 ms (R=50 kΩ), as shown in Fig. 7(a), were applied to the electrostatic film motor, the driving currents $\mathbf{I}$ were obtained, as shown in Fig. 7(b). The real (simulated) slider position x is plotted in Fig. 7(c) using black line. Because the external disturbance was applied to the motor at around t=0.12 s, as shown in Fig. 7(c) using red line, the motor was forced to step out from its synchronous movement.

Fig. 8 shows the signal processing of the proposed method. By removing the switching currents from the driving currents shown in Fig. 7(b), the plateau currents $\mathbf{I}_p$ were obtained, as shown in Fig. 8(a).

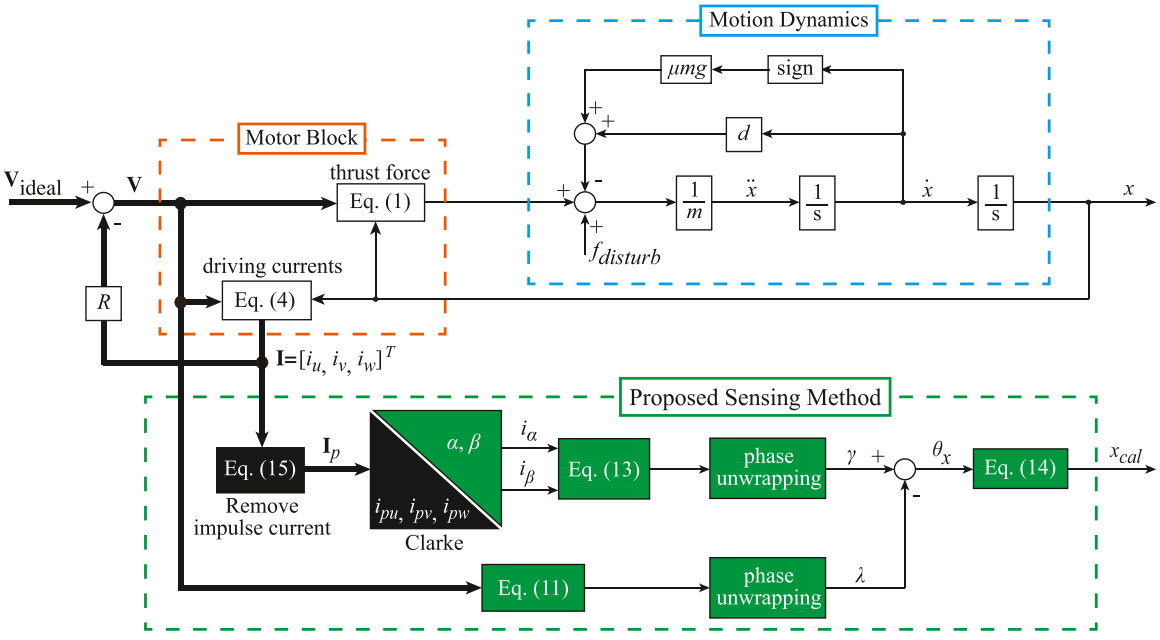


Fig. 6. Block diagram of an electrostatic film motor and the proposed position estimation method. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

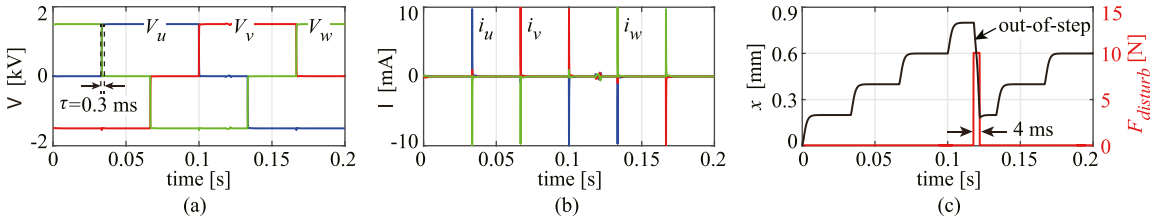


Fig. 7. Power supply with a rise time of 0.3 ms and the real slider position x. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

Although the switching currents were not removed completely, the remained parts were small enough compared to the original impulse. The plateau currents corresponding to out-of-step at t=0.12 s have no loss, as it occurred between two switching times. Then, the plateau currents $\mathbf{I}_p$ were simplified by using the Clarke transformation, as shown in Fig. 8(b). The phase γ was calculated within ($-\pi/2$, $\pi/2$) by using Eq. (13), as shown in Fig. 8(c). The discontinuous points in the phase-wrapped γ were eliminated by using the phase unwrapping method in [49,50], which gives Fig. 8(d). It was confirmed that the phase γ changes with a step of $\pi/3$ as the motor moves synchronously. The phase λ was derived from the applied voltage by using Eq. (11), as shown in Fig. 8(e), which also changes with a step of $\pi/3$. As θ_x is the sum of γ and λ, θ_x changes with a step of $2\pi/3$, which corresponds to 1 electrode pitch. The slider position x_{cal} was calculated by substituting γ and λ into Eq. (14), as shown in Fig. 8(f) using purple line. For comparison, the real slider position x shown in Fig. 7(c) was also plotted in the same figure using black line. Blue line expresses the absolute error ε between x_{cal} and x, which is in the range ($-p$,p). Although some error is observed during the step movement of the slider, the error is eliminated soon after the slider settles in the new position. Also, the slider position during out-of-step was tracked correctly.

Simulations were also carried out under different voltage rise times which were changed indirectly by adjusting resistance R with an interval of 10 kΩ from 0 to 500 kΩ. Some results are shown in Fig. 9. Purple lines are the calculated slider position x_{cal}, while black lines are the real slider position x. When the disturbance was applied to the motor at t=0.12 s, the proposed method tracked the slider position correctly at both synchronous movement and out-of-step, as shown in Fig. 9(a) and (b). When the disturbance was given at around the switching time, the proposed method can only work well when $\tau \leqslant 0.55$ ms ($R \leqslant 110$ kΩ), as shown in Fig. 9(c) and (d). To clarify the reason of the failure, the plateau currents $\mathbf{I}_p$ and the calculated phase γ around out-of-step in Fig. 9(c) and (d) were illustrated in Fig. 10. When $\tau \leqslant 0.55$ ms, the "R-jump" $\Delta\gamma_\tau$ was smaller than $\pi/2$, the phase γ was calculated correctly, as shown in Fig. 10(a). When the "R-jump" $\Delta\gamma_\tau$ became larger than $\pi/2$, as shown in Fig. 10(b), the phase γ could not be obtained correctly from the wrapped γ due to the information loss, and then the position estimation failed with large error. It was confirmed that the proposed position estimation method for the pulse-voltage operation works well when the rise time of the power supply is short enough ("R-jump" $\Delta\gamma_\tau$ satisfy the requirement in Eq. (18)), regardless of when out-of-step occurs.

3.3. Discussion

Although the proposed position estimation method has low accuracy, it is possible to be improved by optimizing the configuration of electrodes. In the theoretical derivation of the method in section II, the capacitance coefficient C_{m2} in the capacitance matrix in Eq. (2) was ignored for simplicity. However, the simulated motor adopted non-zero value for C_{m2}. As a result, an additional component was created in the driving currents, which caused errors and affected the accuracy of the proposed method. On the other hand, smooth motion is generated under sine-voltage operation by using a configuration of zig-zag electrodes, in which the capacitance coefficient C_{m2} does not exist [18]. Therefore, the proposed position estimation has potential to be increased by using the configuration of zig-zag electrodes due to the disappearance of C_{m2}.

The rise time plays different role during synchronous motion and out-of-step. Fig. 8(d) confirmed that the change of the phase γ between two steps is $\pi/3$, which always satisfies the success condition in Eq. (18). Therefore, the proposed position estimation method can not be affected by the rise time of the driving voltages during synchronous movement. On the other hand, the performance for out-of-step is restricted by the rise time. When out-of-step occurred during

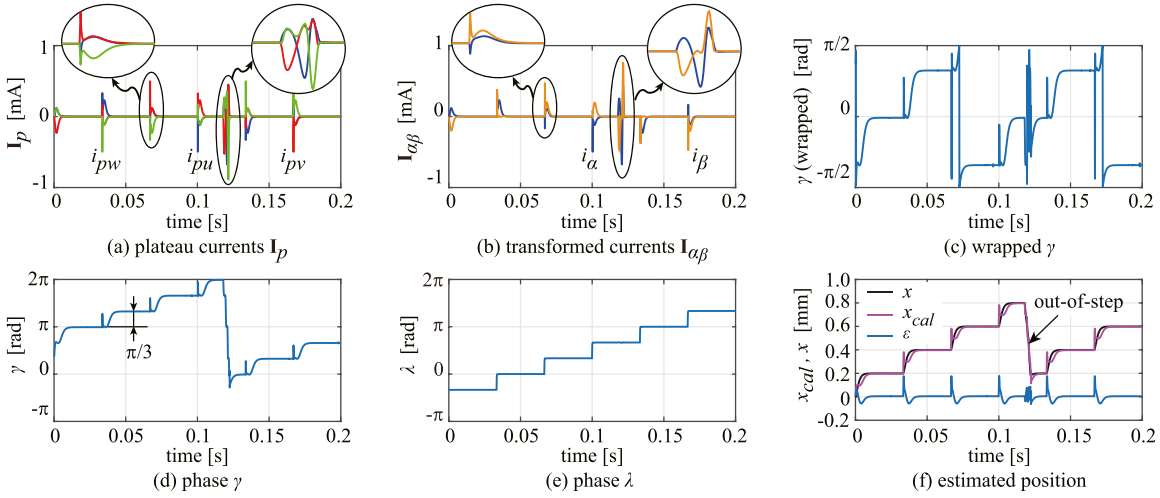


Fig. 8. Signal processing of the proposed position estimation method (simulation, rise time of the pulse voltage is τ=0.3 ms). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

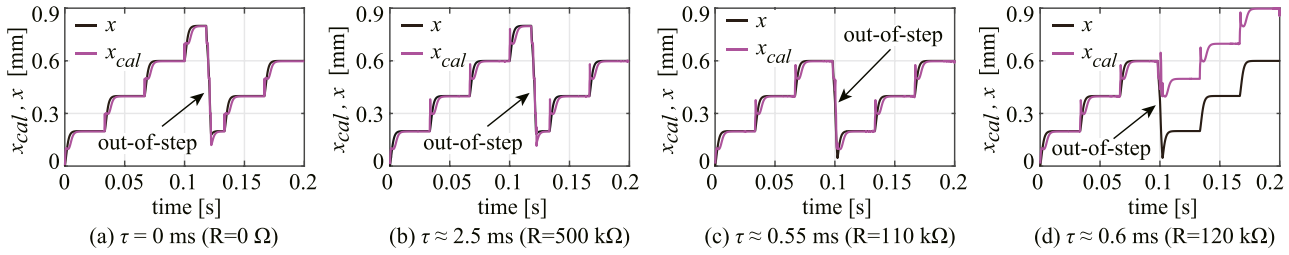


Fig. 9. Estimated displacements under pulse-voltage with different rise times (simulation).

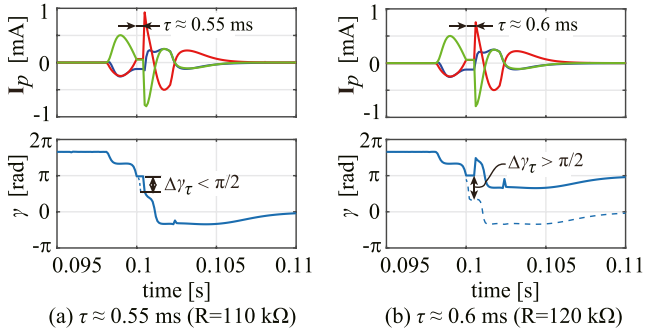


Fig. 10. Remained plateau currents $\mathbf{I}_p$ and calculated phase γ around the out-of-step in Fig. 9(c) and (d) (simulation).

a plateau period, the slider motion can be tracked correctly under any conditions due to no information loss. Meanwhile, if out-of-step occurs around the switching time, the slider position can only be estimated correctly in case of short rise time. Considering the random appearance of out-of-step in practice, it is suggested that power supply with high performance (short rise time) should be utilized in practical applications.

4. Experiment

4.1. Setup

The proposed position estimation method for the pulse-voltage operation was applied to an experimental system, in which the driving currents were measured and processed online. The experiment setup is shown in Fig. 11. The motor prototype is same as the "regular motor" in [45] whose electrode pitch is 200 μm. The electrode films were attached to a plastic plate or a plastic box, and the effective driving electrode area is 110 cm^2. The slider mass including the slider film and the plastic plate was 30 g. Glass beads with a diameter of 20 μm were sprinkled between films to reduce the friction, and the motor was operated in a dielectric fluid (3 M, Flouorinert FC-77) for high performance. Random external disturbances, that were applied to the sliders to force the slider step out, were provided by a human hand.

A 3-phase pulse-voltage was generated by using a digital signal processing system (Dspace, ds1104), and was amplified by a set of high-voltage amplifiers (NF, HVA4321) with high performance (short rise time). The 3-phase pulse-voltage with an amplitude of ± 1.0 kV and a switching frequency of 6 Hz, was applied to the motors, as shown in Fig. 11. A set of current sensors based on voltage dividing circuit was used to measure the driving currents, whose full scale is 0.5 mA with an output gain of 0.05 mA/V. Only two phases of the driving currents were measured as the third phase i_{pw} is not required as shown in Eq. (9). The real slider position was detected by using a laser displacement sensor (Matsushita Electric Works, ANR-1282). The output of the current sensors were acquired by the DSP system for position estimation, whose sampling frequency was 10 kHz.

4.2. Results

The measured driving currents of the electrostatic film motor are shown in Fig. 12(a), while the real slider position x measured by using the laser displacement sensor is shown in Fig. 12(b) using black line. External disturbances from human hand were applied to the motor around t=0 s, t=1.5 s and t=4 s, and the motor was forced to step out from synchronous motion. The estimated slider position x_{cal} was calculated from the driving currents by using the proposed method, whose signal processing will be introduced in the following paragraph, as shown in Fig. 12(b) using purple line. The error ε=x_{cak}−x is plotted in Fig. 12(c). It is confirmed that the error ε is around one electrode pitch p, which was predicted in the simulation.

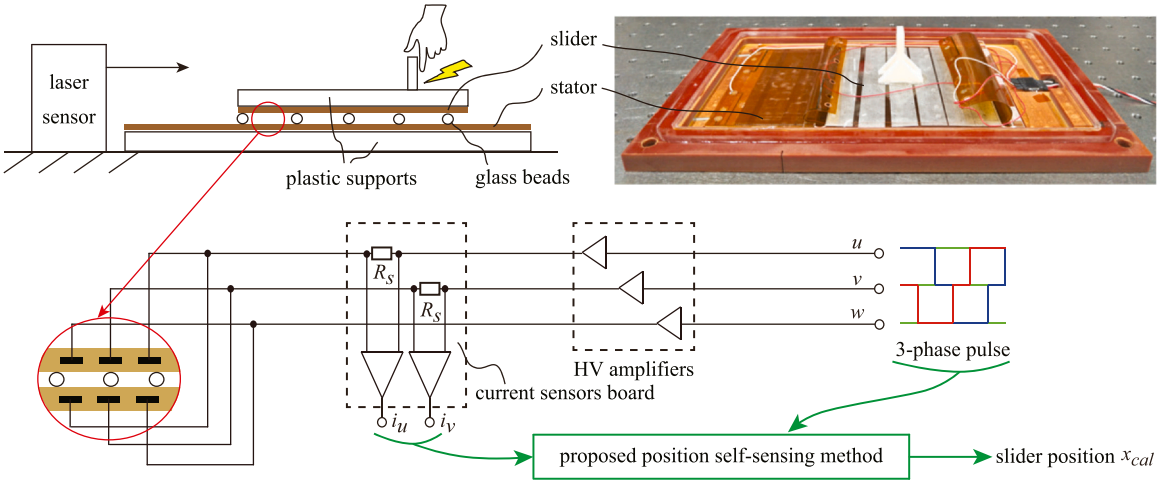


Fig. 11. Experiment setup for verifying the proposed position estimation method for pulse-voltage operation.

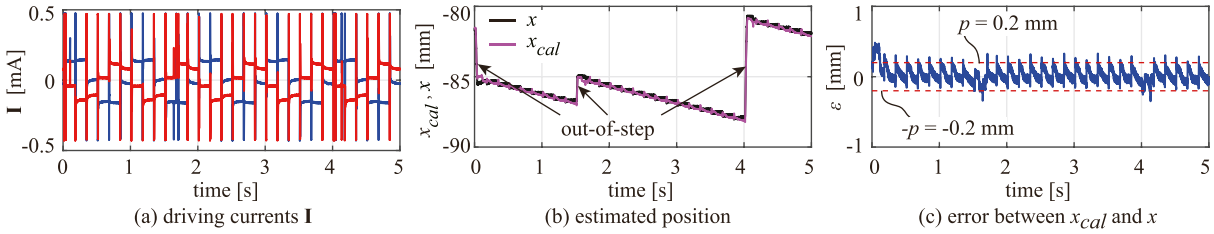


Fig. 12. Driving currents and estimated slider position.

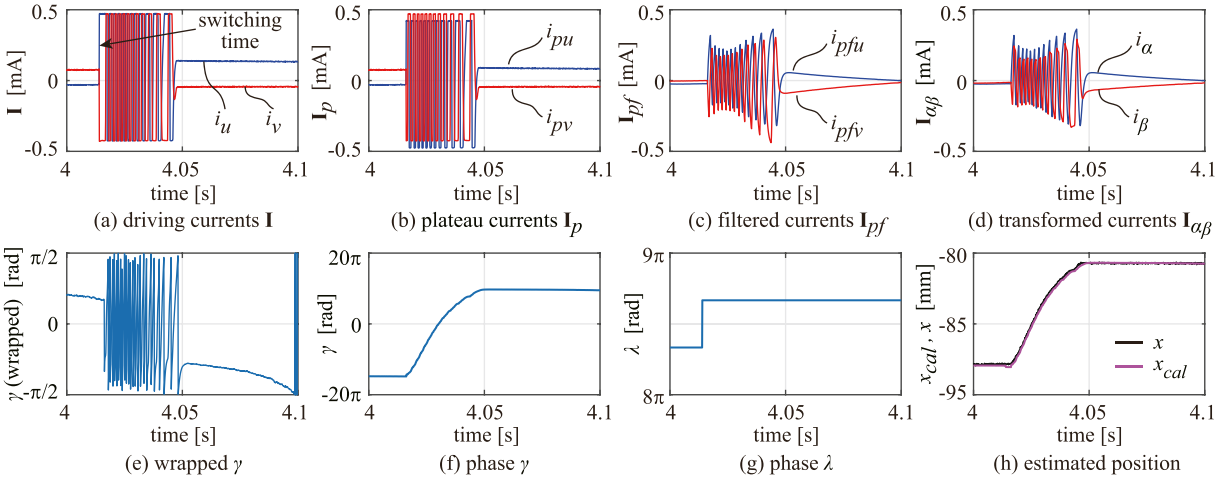


Fig. 13. Signal processing around the third out-of-step in Fig. 12 (rise time of the applied voltages is τ=0.8 ms).

Taking the third out-of-step in Fig. 12(b) for example, the signal processing of the proposed method in experiment was introduced, as shown in Fig. 13. Fig. 13(a) is the driving currents around t=4 s. Although the currents caused by out-of-step occurred around the switching time, most parts were in the plateau period of the pulse-voltage, which is similarly with that in Fig. 8(a). It is observed that the driving currents were out of the range of the current sensors when the slider stepped out. Using the rise time of the applied pulse-voltage τ=0.8 ms, plateau currents $\mathbf{I}_p$ were extracted by removing the switching currents using Eq. (15), as shown in Fig. 13(b). Different with the signal processing in simulation, filters were used to remove the noise and DC offset from the measured currents. Considering the influence of the impulse currents, the filters were applied to the plateau currents $\mathbf{I}_p$ after removing the switching currents. A 2nd-order Butterworth LPF with a cut-off frequency of 200 Hz was used to remove the noise and a 2nd-order Butterworth HPF with a cutoff frequency of 2 Hz was used to remove the DC offset. The filtered currents $\mathbf{I}_{pf}$ is shown in Fig. 13(c). After Clarke transformation, the filtered currents $\mathbf{I}_{pf}$ were transferred into $\mathbf{I}_{\alpha\beta}$, as shown in Fig. 13(d). By using Eq. (13), the phase γ was calculated in the range $(-\pi/2,\pi/2)$, as shown in Fig. 13(e). And then an phase-unwrapping method was used to eliminate the discontinuity, which gives Fig. 13(f). The phase λ was calculated according to Eq. (11) by using the generated 3-phase pulse-voltage in the digital signal processing system, which is shown in Fig. 13(g). Then, the slider position x_{cal} was calculated by using Eq. (14), as shown in Fig. 13(h) using purple line. For comparison, the real slider position x was also plotted in the same figure using black line. It is confirmed that the phase γ and the slider position x_{cal} can be calculated correctly around out-of-step as the corresponding plateau current was remained, which coincide with the simulation results. Although the driving currents exceeded the range of the current sensors during the out-of-step, it did not have negative impact to the position estimation method, which is because the calculation depends on the phase relation between currents rather than their values.

To verify the feasibility of the proposed position estimation method, additional experiments have been done under other conditions: "blocked by a wall" and "variable-frequency drive". The results are shown in Fig. 14, in which purple lines are the calculated slider position x_{cal} and black lines express the real slider position x.

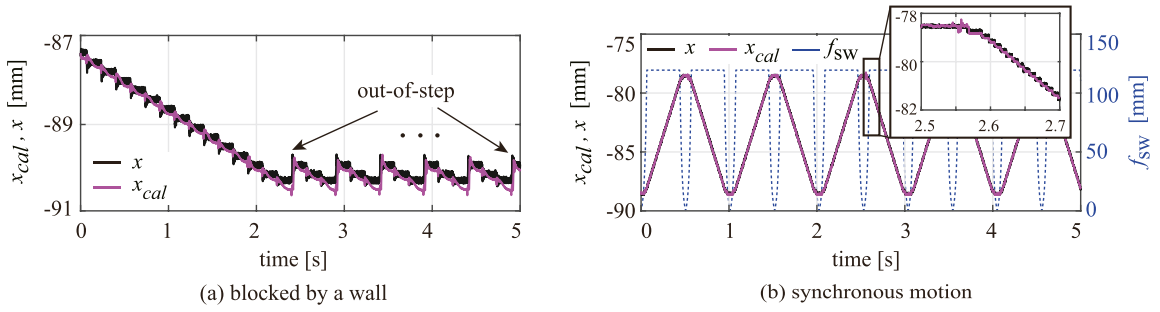


Fig. 14. Position estimation by using the proposed method under various conditions.

In case of "blocked by a wall", the motor was forced to step out when it hit a rigid block, which is same with the experimental condition in [45]. The result is shown in Fig. 14(a), in which the motor was forced to step out when it hit the wall, around 2.4 s, 2.9 s,3.4 s, 3.9 s,4.4 s, 4.9 s. In case of "variable-frequency drive", the motor moved in a reciprocating motion without any external disturbance, as shown in Fig. 14(b). The switching frequency f_{sw} of the pulse-voltage was increased from 0 Hz to 120 Hz at the time of starting and was reduced from 120 Hz to 0 Hz for braking, which is indicated by using the blue dashed line. It is confirmed that the proposed method can work well under various conditions.

Although the proposed method worked fine with the prototype motor, it was found that the method can fail for another motor prototype if the same sensing parameters are utilized. Even if the amplifiers are the same, use of different motor can result in different rise time τ, which can cause tracking error especially for out-of-step motions. This means that even if the same voltage amplifiers are utilized, the rise time needs to be fine tuned for individual motors.

5. Conclusions

The position estimation method for the pulse-voltage operation was proposed in this work, in which the impulse currents at the switching time of the pulse-voltage are removed directly. The rise time of the pulse-voltage was determined as the removal period of the impulse currents. Simulation and experiment results confirmed that the proposed method can track the slider position correctly during both synchronous movement and "out-of-step" when the removal time is short enough, and failed when the removal time became longer due to the information loss. The experiments under various conditions verified the effectiveness and the feasibility of the proposed method by using a real regular motor. Although the resolution of the proposed method is not high (around one electrode pitch), it is helpful to the synchronous motors for recovering position from out-of-step and increasing the fault tolerance with a considerably simpler and low-cost structure. As the accuracy of the proposed method is possible to be improved by using optimized electrode configuration, the future research should focus on the challenges existing in the optimization of electrode configuration under pulse-voltage operation.

CRediT authorship contribution statement

Guangwei Zhang: Conceptualization, Methodology, Software, Investigation, Writing – original draft. **Akio Yamamoto:** Conceptualization, Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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